Daejeon, Korea
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Last modified : June 1, 2026

Hyunmo Sung

https://programelot.github.io
GitHub: programelot
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Skills

Programming	C, C++, C#, CUDA, Python, PAPI, CMake, LLVM
Communication	Korean (native), English
Other	Unity, Visual studio code, Github, Linux(Ubuntu)

Work Experience

Engineer — **March 2026 -**
Samsung Electronics — *Suwon, Korea*

Researcher — **February 2020 - February 2023**
ELC(Embedded Systems Languages and Compilers Lab) — *Yonsei University, Seoul, Korea*

- Researched about the profile-guided-optimization.
- Implemented and evaluated the performance of the bloom filter using CUDA unified memory.
- Evaluated the performance of lazy parallel kronecker algebra on the modern GPU T4.
- Joint Research Project with DS Division of Samsung Inc. through Yonsei-Samsung Semiconductor Research Center (YSSRC) Program.

Internship — **July 2019 - February 2020**
ELC(Embedded Systems Languages and Compilers Lab) — *Yonsei University, Seoul, Korea*

- Researched about the kronecker algebra to detect the deadlock of the program.

Internship — **December 2014 - February 2015**
RiseGroup — *Dongguk University, Seoul, Korea*

- Worked as a subprogrammer. Developed a serious game to prevent school violence.

Project

Revist the Supernodal Floyd Warshall algorithm — **November 2023 - December 2023**
Side project

- Implemented the Supernodal Floyd-Warshall algorithm from the paper https://dl.acm.org/doi/10.1145/3332466.3374533.
- 78 times fasters than a Floyd-Warshall algorithm with a single thread.
- 17 times fasters than a Floyd-Warshall algorithm with multiple threads with other optimization techniques.
- Utilized entire threads with OpenMP.
- Source is open on the github repository https://github.com/programelot/Supernodal-Floyd-Warshall.

Matrix multiplication on GPU — **July 2022 - September 2022**
Side project

- Developed 11 matrix multiplication algorithms on CPU and GPU.
- Estimated and compared performance of each algorithms.
- Utilized the shared memory on GPU.
- Implemented Strassen's algorithm and Winograd's algorithm.
- Source is open on the github repository https://github.com/programelot/MatrixMultiplication.

BackRomii **July 2022**
Indie Game Development

- Developed an escape game from auto generated maze with first person view.
- - Used recursive divison method to generate maze.
- Optimized the game by developing an off culling algorithm.
- Developed a path finding algorithm.
- Developed in two weeks like game jam.
- Game has been released on the web. https://aintmos.itch.io/backromii
- 1162 views and 292 downloads happened.

Education

Computer science **March 2020 - February 2023**
Yonsei university, Seoul, Korea
Master, Gradutated

Main courses: Multicore computing topics, Introduction to approximation algorithms

Computer science **March 2016 - February 2020**
Yonsei university, Seoul, Korea
Bachelor, Gradutated

Main courses: Algorithm analysis, Computer graphics, Compiler design, Multicore programming fundamentals

Multimedia engineering **March 2014 - February 2016**
Dongguk university, Seoul, Korea
Bachelor, Drop out for transfer to other university

Main courses: Multimedia Data Structures, Internet Programming, Multimedia Programming

Publication

Profile-guided Orchestration of Computations for CPU and PIM Cores **2023**
Hyunmo Sung
Yonsei university, thesis.

Performance Evaluation of GPU-based Bloom Filters Using CUDA Unified Memory **2022**
***Hyunmo Sung**, and Bernd Burgstaller*
Korea Software Congress 2022 (한국정보과학회 학술발표논문집 2022): 45-47.

Lazy Evaluation of Kronecker Algebra Operations on the Tesla T4 GPU **2020**
*Ham, Seokhwan, **Hyunmo Sung**, Shinhyung Yang, and Bernd Burgstaller*
Korea Computer Congress 2020 (한국정보과학회 학술발표논문집 2020): 44-46.

Patent

Offloading methodology for utilizing Processing-In-Memory and the machine for it
(프로세스 인 메모리의 활용을 위한 오프로드 처리 방법 및 그를 위한 장치)
*Bernd Burgstaller, **Hyunmo Sung**, Seongho Jeong, Shinhyung Yang, Jayhwan Lee, and Jiun Jung.*
Application No. 10-2022-0162906, Nov 29 2022.

Awards

- **Capstone project 1st place (졸업 작품 최우수상)** **December 06, 2019**
  Lazy Parallel Kronecker Algebra, Yonsei University, Seoul, Korea

- **Capstone project 1st place (졸업 작품 최우수상)** **May 13, 2019**
  Projection-Based AR Evacuation Simulator using Kinect for Windows V2, Yonsei University, Seoul, Korea
- **Honored Student Prize (학기 우등생)** **July 09, 2015**
  Dongguk University, Seoul, Korea
- **Honored Student Prize (학기 우등생)** **January 09, 2015**
  Dongguk University, Seoul, Korea
- **Honored Student Prize (학기 우등생)** **July 07, 2014**
  Dongguk University, Seoul, Korea